Exercise 1 – ML Basics

Introduction to Machine Learning

Exercise 1: Types of learning problems

Learning goals

1. Distinguish supervised from unsupervised tasks (and both from problems where nothing is learned from data)
2. Distinguish regression from classification tasks
3. Identify prediction vs. explanation goals

Below you find short descriptions of five data-related problems. For each scenario, answer the following questions and justify your answers briefly:

- Is this a machine learning problem at all? If so, is it supervised or unsupervised?
- If it is supervised: is it a regression or a classification task?
- Is the primary goal learning to predict or learning to explain?
- For supervised scenarios: what is the target variable (including a sensible target space $\mathcal{Y}$), and what could be suitable features?

a) You work at a second-hand car dealer and want to acquire for-sale vehicles at reasonable prices. To support this, you plan to build a model that predicts a car's adequate market price (in EUR) from its properties, such as age and mileage, using records of past sales.

> **Solution**
>
> Since past sale prices are recorded, we have labeled data, so this is a **supervised** ML problem. The target (price in EUR) is continuous, which makes this a **regression** task with $\mathcal{Y} = \mathbb{R}$ (in practice non-negative). The goal is **prediction**: you want accurate price estimates for new cars to spot good deals; still, the explanatory contribution of individual features (e.g., how much an extra year of age lowers the price) may be of secondary interest. Features: age, mileage, brand, ... We formalize and build a full learner for this problem in Exercise 2.

b) A team of epidemiologists has examined several thousand children, recording for each child whether they have been diagnosed with asthma, along with the average particulate-matter concentration at their home, their age, and whether a parent smokes. They want to quantify how strongly particulate matter affects the risk of a child being asthmatic, accounting for the other influences.

Solution

Since each child's asthma diagnosis is recorded, we have an observed target (a label), so this is a **supervised** ML problem (a very classical statistical modeling one). The target is binary (asthmatic yes/no), which makes this a **classification** task with $\mathcal{Y} = \{\text{asthma}, \text{no asthma}\}$. The goal is **explanation**: the researchers want to quantify the effect of particulate matter on asthma *risk* – as one would with an interpretable model such as logistic regression, reading off effect estimates – not to predict the diagnosis of any individual child. Features: particulate-matter concentration plus the controls (age, parental smoking, ...).

c) A music streaming service wants to organize its catalog into groups of similar-sounding songs, based on audio characteristics such as tempo, energy, and spectral properties. There are no predefined categories the songs should be sorted into.

Solution

This is an ML problem, but an **unsupervised** one: there is no labeled target we could learn to predict – we are looking for structure (groups of similar observations) within the inputs, which is a **clustering** task. Consequently, the regression/classification and predict/explain distinctions do not apply (they are concepts of supervised learning).

d) Based on records of past loans with known repayment outcomes, a bank wants to decide on new loan applications by estimating each applicant's probability of repaying. Regulation requires the bank to be able to justify to applicants why they were rejected.

Solution

This is a **supervised classification** problem: from past loans with known outcomes, the bank learns to classify new applications, with $\mathcal{Y} = \{\text{repaid}, \text{default}\}$. Features could be income, employment status, existing debt, or credit history. The interesting part is the goal: the primary purpose is **prediction** (accurate decisions on new applications), but the regulatory requirement adds an **explanation** component – the bank must be able to state *why* an application was rejected. This is a typical real-world case where both goals matter and the choice of model class must reflect that (a topic we will return to throughout the lecture).

e) A smoke detector should sound an alarm as soon as the measured particle concentration in the air exceeds a threshold defined in a technical standard.

Solution

This is **not a machine learning problem**: the decision rule (threshold comparison) is fully specified in advance and not learned from data. What decides whether something is ML is precisely this – whether the model is obtained by learning from data – and not what the final decision rule happens to look like. Note that a related ML problem *could* be constructed, e.g., learning from sensor data and past incidents when an alarm is warranted, but as stated, there is neither training data nor anything to optimize.

Exercise 2: Car price prediction

Learning goals

1) Translate a real-world problem into ML concepts
2) Use proper mathematical notation for those concepts
3) Decompose a learner into hypothesis space, risk, and optimization

In Exercise 1 a), you characterized the used-car pricing problem as a supervised regression task: you work at a second-hand car dealer and want a model that predicts adequate market prices (in EUR) from vehicles' properties. Let us now develop a complete learner for it.

a) How would you set up your data? Name potential features along with their respective data type and state the target variable.

Solution

Target variable and potential features:

Variable	**Role**	**Data type**
Price in EUR	Target	Numeric
Age in days	Feature	Numeric
Mileage in km	Feature	Numeric
Brand	Feature	Categorical
Accident-free y/n	Feature	Binary
...	...	...

b) Assume now that you have data on vehicles' age (days), mileage (km), and price (EUR). Explicitly define the feature space $\mathcal{X}$ and target space $\mathcal{Y}$.

Solution

Let x_1 and x_2 measure age and mileage, respectively. Both features and target are numeric and (quasi-) continuous. It is also reasonable to assume non-negativity for the features, such that we obtain $\mathcal{X} = (\mathbb{R}_0^+)^2$, with $\mathbf{x}^{(i)} = (x_1^{(i)}, x_2^{(i)})^\top \in \mathcal{X}$ for $i = 1, 2, \dots, n$ observations. As the standard LM does not impose any restrictions on the target, we have $\mathcal{Y} = \mathbb{R}$, though we would probably discard negative predictions in practice.

c) You choose to use a linear model (LM) for this task. The LM models the target as a linear function of the features with Gaussian error term.

State the hypothesis space for the corresponding model class. For this, assume the parameter vector θ to include the intercept coefficient.

Solution

We can write the hypothesis space as:

$$\mathcal{H} = \{f(\mathbf{x} \mid \theta) = \theta^\top \mathbf{x} \mid \theta \in \mathbb{R}^3\} = \{f(\mathbf{x} \mid \theta) = \theta_0 + \theta_1 x_1 + \theta_2 x_2 \mid (\theta_0, \theta_1, \theta_2) \in \mathbb{R}^3\}.$$

Note the **slight abuse of notation** here: in the lecture, we first define θ to only consist of the feature coefficients, with $\mathbf{x}$ likewise being the plain feature vector. For the sake of simplicity, however, it is more convenient to append the intercept coefficient to the vector of feature coefficients. This does not change our model formulation, but we have to keep in mind that it implicitly entails adding an element 1 at the first position of each vector.

d) Which parameters need to be learned? Define the corresponding parameter space Θ.

Solution

The parameter space is included in the definition of the hypothesis space and in this case given by $\Theta = \mathbb{R}^3$.

e) State the loss function for the i-th observation using $L2$ loss.

Solution

Loss function for the i-th observation: $L\left(y^{(i)}, f\left(\mathbf{x}^{(i)} \mid \theta\right)\right) = \left(y^{(i)} - \theta^\top \mathbf{x}^{(i)}\right)^2$.

f) Now you need to find the best parameters, and hence the best model, via empirical risk minimization. State the corresponding optimization problem formally.

Solution

In order to find the optimal $\hat{\theta}$, we need to solve the following minimization problem:

$$\hat{\theta} = \arg\min_{\theta \in \Theta} \mathcal{R}_{\text{emp}}(\theta) = \arg\min_{\theta \in \Theta} \left(\sum_{i=1}^{n} \left(y^{(i)} - \theta^\top \mathbf{x}^{(i)}\right)^2 \right)$$

For the linear model, this problem can even be solved analytically, yielding the familiar least-squares estimator (lecture group A derives it in Exercise 4).

g) Finally, take a step back: the lecture summarizes supervised learning as *Learning = Hypothesis Space + Risk + Optimization*. Match your results from c) through f) to these three components.

Solution

- **Hypothesis space:** the set of admissible models we restrict ourselves to a priori – here, the space of linear functions from c), parametrized over $\Theta = \mathbb{R}^3$ from d).
- **Risk:** the criterion that assigns each candidate model a quality score on the training data – here, the empirical risk $\mathcal{R}_{\text{emp}}(\theta)$, obtained by summing the pointwise $L2$ losses from e) over all observations.
- **Optimization:** the strategy to find the model with minimal risk – here, solving the minimization problem from f).

Together, these three components fully specify our learner: it takes the training data and returns the empirically best linear model $\hat{\theta}$.

Congratulations, you just designed your first machine learning project!

Exercise 3: Comparing models via empirical risk

The whole exercise is only for lecture group B!

Learning goals

1. Compute pointwise losses and empirical risk by hand
2. Understand how the choice of loss function affects model comparison
3. Explain how risk minimization selects the optimal model

You are back at the second-hand car dealer from Exercise 2. To get a feeling for how loss functions work, you study a small data set of $n = 5$ used cars of one popular vehicle model, with age (in years) as the only feature and price (in 1000 EUR) as the target:

Car i	Age $x^{(i)}$	Price $y^{(i)}$
1	1	16
2	2	14
3	4	10
4	6	6

Car i	Age $x^{(i)}$	Price $y^{(i)}$
5	8	10

Car 5 deviates from the general pattern: it is a sought-after classic edition whose price rises again with age.

Two of your colleagues propose two different pricing models:

$$f_1(x) = 18 - 2x, \qquad f_2(x) = 16 - x.$$

a) For both models, compute the predictions $f(x^{(i)})$ for all five cars, as well as the pointwise losses under the $L1$ loss $\left|y^{(i)} - f\left(\mathbf{x}^{(i)}\right)\right|$ and the $L2$ loss $\left(y^{(i)} - f\left(\mathbf{x}^{(i)}\right)\right)^2$.

Solution

In the following table, the $L1$ and $L2$ columns contain the pointwise losses of the model to their left:

Car i	$y^{(i)}$	$f_1(x^{(i)})$	$L1$	$L2$	$f_2(x^{(i)})$	$L1$	$L2$
1	16	16	0	0	15	1	1
2	14	14	0	0	14	0	0
3	10	10	0	0	12	2	4
4	6	6	0	0	10	4	16
5	10	2	8	64	8	2	4

f_1 predicts cars 1–4 perfectly but misses the classic car 5 by a wide margin; f_2 makes moderate errors on almost all cars.

b) Compute the empirical risk $\mathcal{R}_{\text{emp}}(f)$ of both models under both losses. Which model is preferable under $L1$ loss, and which under $L2$ loss?

Solution

Summing the pointwise losses from a):

	$\mathcal{R}_{\text{emp}}(f_1)$	$\mathcal{R}_{\text{emp}}(f_2)$	preferable model
$L1$	$0+0+0+0+8=8$	$1+0+2+4+2=9$	f_1
$L2$	$0+0+0+0+64=64$	$1+0+4+16+4=25$	f_2

So which model is "better" **depends on the loss function** we choose: under $L1$ loss we would pick f_1, under $L2$ loss f_2.

c) Explain why the two loss functions disagree about the better model. What role does car 5 play?

> **Solution**
>
> Car 5 is an **outlier** with respect to the general price pattern. The $L2$ loss squares residuals, so a single large error dominates the empirical risk: f_1's one big mistake ($8^2 = 64$) outweighs all of f_2's moderate errors combined. The $L1$ loss only grows linearly in the residuals and is therefore more **robust** to outliers: it rewards f_1 for fitting the majority of cars perfectly and penalizes the single large error much less severely. The choice of loss is thus a real design decision: do we prefer a model that is very accurate for typical cars but can be far off for unusual ones (f_1 under $L1$), or one that avoids large errors on any car (f_2 under $L2$)?

d) Our two candidate models form a tiny hypothesis space $\mathcal{H} = \{f_1, f_2\}$. Explain in your own words, and state with a formula, how the principle of empirical risk minimization selects the optimal model $\hat{f}$ from a hypothesis space.

> **Solution**
>
> The empirical risk assigns each candidate model a single quality score: the total loss it incurs on the training data. Empirical risk minimization then simply selects the model with the **smallest** score,
>
> $$\mathcal{R}_{\text{emp}}(\hat{f}) = \min_{f \in \mathcal{H}} \mathcal{R}_{\text{emp}}(f),$$
>
> i.e., $\hat{f}$ is the element of $\mathcal{H}$ achieving the minimal empirical risk. For our finite hypothesis space we can find $\hat{f}$ by tabulating the risks exhaustively, as we did in b): under $L1$ loss, $\hat{f} = f_1$; under $L2$ loss, $\hat{f} = f_2$. For the infinitely large hypothesis spaces of real model classes, we instead search over the parameter space – this is exactly the optimization problem from Exercise 2 f).

Exercise 4: Vector calculus and the least-squares estimator

> The whole exercise is only for lecture group A!

> Learning goals
>
> 1. Understand how vector-valued functions work
> 2. Perform calculus on vectors and matrices
> 3. Derive the least-squares estimator via empirical risk minimization

Consider the following function performing matrix-vector multiplication: $f(\mathbf{x}) = \mathbf{A}\mathbf{x}$, where $\mathbf{A} \in \mathbb{R}^{m \times n}$, $\mathbf{x} \in \mathbb{R}^{n \times 1}$.

a) What is the dimension of $f(\mathbf{x})$? Explicitly state the calculation for the i-th component of $f(\mathbf{x})$.

> **Solution**
>
> In computing $\mathbf{A}\mathbf{x}$ we multiply each of the m rows in $\mathbf{A}$ with the sole length-n column in $\mathbf{x}$, leaving us with a column vector $f(\mathbf{x}) \in \mathbb{R}^{m \times 1}$. Thus, we have $f : \mathbb{R}^{n(\times 1)} \to \mathbb{R}^{m(\times 1)}$.
>
> The i-th function component $f_i(\mathbf{x})$ corresponds to multiplying the i-th row of $\mathbf{A}$ with $\mathbf{x}$, amounting to
>
> $$f_i(\mathbf{x}) = \sum_{j=1}^{n} a_{ij} x_j,$$
>
> with a_{ij} the element in the i-row, j-th column of $\mathbf{A}$.

b) Now, consider the gradient (derivative generalized to multivariate functions) $\frac{\mathrm{d}f(\mathbf{x})}{\mathrm{d}\mathbf{x}}$ (a.k.a. $\nabla_{\mathbf{x}} f(\mathbf{x})$).

(i) What is the dimension of $\frac{\mathrm{d}f(\mathbf{x})}{\mathrm{d}\mathbf{x}}$?

> **Solution**
>
> The gradient is the row vector[a] of partial derivatives, i.e., the derivatives of f w.r.t. each dimension of $\mathbf{x}$:
>
> $$\frac{\mathrm{d}f(\mathbf{x})}{\mathrm{d}\mathbf{x}} = \begin{pmatrix} \frac{\partial f(\mathbf{x})}{\partial x_1} & \dots & \frac{\partial f(\mathbf{x})}{\partial x_n} \end{pmatrix}.$$
>
> Now, since f is a vector-valued function, each component is itself a vector of length m. Therefore, we have $\frac{\mathrm{d}f(\mathbf{x})}{\mathrm{d}\mathbf{x}} \in \mathbb{R}^{m \times n}$, given by the collection of all partial derivatives of each function component:
>
> $$\frac{\mathrm{d}f(\mathbf{x})}{\mathrm{d}\mathbf{x}} = \begin{pmatrix} \frac{\partial f_1(\mathbf{x})}{\partial x_1} & \dots & \frac{\partial f_1(\mathbf{x})}{\partial x_n} \\ \vdots & & \vdots \\ \frac{\partial f_m(\mathbf{x})}{\partial x_1} & \dots & \frac{\partial f_m(\mathbf{x})}{\partial x_n} \end{pmatrix}$$
>
> This matrix is also called the *Jacobian* of f.
>
> [a] Pertaining to one of two conventions; we use the *numerator layout* here (transposed version = *denominator layout*).

(ii) Compute $\frac{\mathrm{d}f(\mathbf{x})}{\mathrm{d}\mathbf{x}}$.

Solution

We have

$$\frac{\partial f_i(\mathbf{x})}{\partial x_j} = \frac{\partial\left(\sum_{j=1}^{n} a_{ij} x_j\right)}{\partial x_j} = a_{ij}.$$

Doing this for every element yields

$$\begin{pmatrix} a_{11} & \cdots & a_{1n} \\ \vdots & & \vdots \\ a_{m1} & \cdots & a_{mn} \end{pmatrix},$$

and we have $\frac{\mathrm{d} f(\mathbf{x})}{\mathrm{d}\mathbf{x}} = \frac{\mathrm{d}\mathbf{A}\mathbf{x}}{\mathrm{d}\mathbf{x}} = \mathbf{A}$.

c) Let's put these rules to use and derive the least-squares estimator you invoked in Exercise 2 f). Recall the ERM problem for the linear model with $L2$ loss:

$$\hat{\theta} = \arg\min_{\theta \in \Theta} \mathcal{R}_{\text{emp}}(\theta) = \arg\min_{\theta \in \Theta} \sum_{i=1}^{n} \left(y^{(i)} - \theta^\top \mathbf{x}^{(i)}\right)^2,$$

where, as in Exercise 2, the first entry of each $\mathbf{x}^{(i)}$ is set to 1, so that $\theta \in \mathbb{R}^{p+1}$ contains the intercept. Collect the feature vectors as rows of the design matrix $\mathbf{X} = \left(\mathbf{x}^{(1)}, \dots, \mathbf{x}^{(n)}\right)^\top \in \mathbb{R}^{n \times (p+1)}$ and the labels in $\mathbf{y} = \left(y^{(1)}, \dots, y^{(n)}\right)^\top$.

(i) Show that the empirical risk can be written as $\mathcal{R}_{\text{emp}}(\theta) = (\mathbf{y} - \mathbf{X}\theta)^\top (\mathbf{y} - \mathbf{X}\theta)$.

Solution

The i-th entry of the vector $\mathbf{X}\theta$ is the product of the i-th row of $\mathbf{X}$ with θ, i.e., $\left(\mathbf{x}^{(i)}\right)^\top \theta = \theta^\top \mathbf{x}^{(i)}$. The residual vector $\mathbf{y} - \mathbf{X}\theta$ therefore has entries $y^{(i)} - \theta^\top \mathbf{x}^{(i)}$, and the inner product of a vector with itself sums up its squared entries:

$$(\mathbf{y} - \mathbf{X}\theta)^\top (\mathbf{y} - \mathbf{X}\theta) = \sum_{i=1}^{n} \left(y^{(i)} - \theta^\top \mathbf{x}^{(i)}\right)^2 = \mathcal{R}_{\text{emp}}(\theta).$$

(ii) Compute the gradient $\frac{\mathrm{d}\mathcal{R}_{\text{emp}}(\theta)}{\mathrm{d}\theta}$.

Hint: Multiply out the risk first. For the quadratic term, you can use the rule $\frac{\mathrm{d}\theta^\top \mathbf{A}\theta}{\mathrm{d}\theta} = \theta^\top \left(\mathbf{A} + \mathbf{A}^\top\right)$ (try to prove it with the product rule).

Solution

Multiplying out yields

$$
\begin{aligned}
\mathcal{R}_{\text{emp}}(\theta) &= \mathbf{y}^\top\mathbf{y} - \mathbf{y}^\top\mathbf{X}\theta - \theta^\top\mathbf{X}^\top\mathbf{y} + \theta^\top\mathbf{X}^\top\mathbf{X}\theta \\
&= \mathbf{y}^\top\mathbf{y} - 2\mathbf{y}^\top\mathbf{X}\theta + \theta^\top\mathbf{X}^\top\mathbf{X}\theta,
\end{aligned}
$$

where we use that $\theta^\top\mathbf{X}^\top\mathbf{y}$ is a scalar and thus equal to its transpose $\mathbf{y}^\top\mathbf{X}\theta$. Now we differentiate term by term, sticking to the numerator layout from b), where gradients of scalar-valued functions are row vectors:

- $\mathbf{y}^\top\mathbf{y}$ does not depend on θ, so its derivative is $\mathbf{0}$.
- $\theta \mapsto \mathbf{y}^\top\mathbf{X}\theta$ is exactly the situation of a)/b) with matrix $\mathbf{A} = \mathbf{y}^\top\mathbf{X} \in \mathbb{R}^{1\times(p+1)}$, so its derivative is $\mathbf{y}^\top\mathbf{X}$.
- For the quadratic term, the hint rule with the *symmetric* matrix $\mathbf{A} = \mathbf{X}^\top\mathbf{X}$ gives $\theta^\top\left(\mathbf{A} + \mathbf{A}^\top\right) = 2\theta^\top\mathbf{X}^\top\mathbf{X}$.

Putting everything together:

$$
\frac{\mathrm{d}\mathcal{R}_{\text{emp}}(\theta)}{\mathrm{d}\theta} = -2\mathbf{y}^\top\mathbf{X} + 2\theta^\top\mathbf{X}^\top\mathbf{X}.
$$

Proof of the hint rule: for $h(\theta) = u(\theta)^\top v(\theta)$ with vector-valued u, v, the product rule reads $\frac{\mathrm{d}h}{\mathrm{d}\theta} = u^\top\frac{\mathrm{d}v}{\mathrm{d}\theta} + v^\top\frac{\mathrm{d}u}{\mathrm{d}\theta}$. With $u = \theta$ and $v = \mathbf{A}\theta$, using b), we get $\theta^\top\mathbf{A} + (\mathbf{A}\theta)^\top\mathbf{I} = \theta^\top\left(\mathbf{A} + \mathbf{A}^\top\right)$.

(iii) Set the gradient to zero to derive the least-squares estimator $\hat{\theta}$. Which assumption about $\mathbf{X}$ is needed for this solution to exist?

Solution

Setting the gradient to zero and transposing both sides yields the so-called **normal equations**:

$$
\begin{aligned}
&-2\mathbf{y}^\top\mathbf{X} + 2\theta^\top\mathbf{X}^\top\mathbf{X} = \mathbf{0} \\
\Leftrightarrow\quad &\mathbf{X}^\top\mathbf{X}\theta = \mathbf{X}^\top\mathbf{y}.
\end{aligned}
$$

If $\mathbf{X}^\top\mathbf{X}$ is invertible, we can solve for θ and obtain the familiar least-squares estimator

$$\hat{\theta} = \left(\mathbf{X}^\top \mathbf{X}\right)^{-1} \mathbf{X}^\top \mathbf{y}.$$

$\mathbf{X}^\top \mathbf{X} \in \mathbb{R}^{(p+1)\times(p+1)}$ is invertible iff $\mathbf{X}$ has full column rank $p+1$, which requires $n \geq p+1$ observations and no linearly dependent columns (e.g., no feature may be a linear combination of the others). The stationary point is indeed a minimum: the Hessian $2\mathbf{X}^\top \mathbf{X}$ is positive semi-definite everywhere (positive definite under full column rank), so $\mathcal{R}_{\text{emp}}(\theta)$ is a convex function of θ.

Alternative: componentwise derivation

Instead of matrix calculus, we can differentiate the empirical risk w.r.t. each component θ_j separately, using the chain rule:

$$\frac{\partial \mathcal{R}_{\text{emp}}(\theta)}{\partial \theta_j} = \sum_{i=1}^{n} 2\left(y^{(i)} - \theta^\top \mathbf{x}^{(i)}\right) \cdot \left(-x_j^{(i)}\right) = -2\sum_{i=1}^{n} x_j^{(i)} \left(y^{(i)} - \theta^\top \mathbf{x}^{(i)}\right).$$

Setting all $p+1$ partial derivatives to zero gives, for each j:

$$\sum_{i=1}^{n} x_j^{(i)} y^{(i)} = \sum_{i=1}^{n} x_j^{(i)} \left(\mathbf{x}^{(i)}\right)^\top \theta.$$

The left-hand sides are exactly the entries of $\mathbf{X}^\top \mathbf{y}$, and the right-hand sides the entries of $\mathbf{X}^\top \mathbf{X}\theta$ – so we arrive at the same normal equations, one coordinate at a time. Both routes are valid; the matrix route is quicker and less error-prone once the calculus rules from a) and b) are familiar.

Had trouble with this exercise?

- For the upcoming contents, you need to be able to handle **matrix-valued computations** (multiplication, transposition etc.) and also matrix-valued **calculus**.
- For more explanations and exercises, including a useful collection of rules for calculus, we recommend the book "Mathematics for Machine Learning" (https://mml-book.github.io/book/mml-book.pdf).